Predator success for torque-based prey interactions and avoidance

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Abstract

Understanding the dynamics between predators and prey has been a fundamental pursuit in ecology and population dynamics. These interactions play a crucial role in shaping ecosystems, influencing biodiversity, and even impacting human activities such as agriculture and conservation efforts. This thesis proposes an extended model of Active Brownian Particles (ABPs) to study predator-prey systems, with a specific focus on how the collective behavior of prey affects predation success. The model simulates particles in a two-dimensional space, incorporating non-reciprocal torque interactions and repulsive steric forces derived from the Weeks-Chandler-Andersen (WCA) potential. Prey particles experience torques that lead to alignment and cohesion from other prey particles, in addition to repulsive torque from the predator.

Extensive simulations varying prey-predator interaction radii R_{p} and prey-predator interaction torque strengths T_0, as well as prey-prey interaction radii R_{pp} and prey-prey interaction torque strengths T_{A}, reveal complex behaviors and dependencies. In the absence of a predator, prey particles exhibit four distinct behavioral states: cloud-like, multiple loose groups, multiple cohesive groups, and one cohesive group. These states transition based on the prey-prey interaction parameters, significantly influencing predator success rates.

The results show that predator success is low when prey exhibit cloud-like, random movement, allowing individual evasion. Predator success increases with prey group cohesiveness, as cohesive groups are more easily targeted and lead to single large eating events. The complex interplay between prey-prey and prey-predator interactions produces varied outcomes, with specific prey cohesion parameters influencing predator success for cohesive prey groups, underscoring the importance of collective prey behavior in predator evasion strategies. Overall, this model provides insights into the dynamics of predator-prey interactions, highlighting the critical role of collective behavior in prey survival and offering potential applications for understanding real-world ecological systems.

Zusammenfassung

Das Verständnis der Dynamik zwischen Räubern und Beutetieren ist ein grundlegendes Bestreben in der Ökologie und Populationsdynamik. Diese Interaktionen spielen eine entscheidende Rolle bei der Gestaltung von Ökosystemen, beeinflussen die Biodiversität und haben sogar Auswirkungen auf menschliche Aktivitäten wie Landwirtschaft und Naturschutzbemühungen. Diese Arbeit schlägt ein erweitertes Modell von Aktiven Brown'schen Partikeln (ABPs) vor, um Räuber-Beute-Systeme zu untersuchen, wobei der Schwerpunkt darauf liegt, wie das kollektive Verhalten der Beute den Jagderfolg beeinflusst. Das Modell simuliert Partikel in einem zweidimensionalen Raum und integriert nicht-reziproke Drehmoment-Interaktionen und abstoßende sterische Kräfte, die sich aus dem Weeks-Chandler-Andersen (WCA) Potential ableiten. Beutepartikel erfahren Drehmomente, die zu Ausrichtung und Kohäsion mit anderen Beutepartikeln führen, zusätzlich zu abstoßenden Drehmomenten vom Räuber.

Umfangreiche Simulationen mit variierenden Beute-Räuber-Interaktionsradien R_p und Beute-Räuber-Interaktionsdrehmomentstärken T_0, sowie Beute-Beute-Interaktionsradien R_pp und Beute-Beute Interaktionsdrehmomentstärken T_A, zeigen komplexe Verhaltensweisen und Abhängigkeiten. In Abwesenheit eines Räubers zeigen die Beutepartikel vier unterschiedliche Verhaltenszustände: wolkenartig, mehrere lose Gruppen, mehrere kohäsive Gruppen und eine kohäsive Gruppe. Diese Zustände wechseln basierend auf den Beute-Beute Interaktionsparametern und beeinflussen erheblich die Jagderfolge der Räuber.

Die Ergebnisse zeigen, dass der Jagderfolg der Räuber gering ist, wenn die Beute ein wolkenartiges, zufälliges Bewegungsmuster aufweisen, das eine individuelle Flucht ermöglicht. Der Jagderfolg der Räuber nimmt mit der Kohäsion der Beutegruppen zu, da kohäsive Gruppen leichter anvisiert werden können und zu großen, einmaligen Fressereignissen führen. Das komplexe Zusammenspiel zwischen Beute-Beute- und Beute-Räuber-Interaktionen führt zu unterschiedlichen Ergebnissen, wobei spezifische Kohäsionsparameter der Beute den Jagderfolg für kohäsive Beutegruppen beeinflussen, was die Bedeutung des kollektiven Verhaltens der Beute in Fluchtstrategien unterstreicht. Insgesamt liefert dieses Modell Einblicke in die Dynamik von Räuber-Beute-Interaktionen, hebt die entscheidende Rolle des kollektiven Verhaltens für das Überleben der Beute hervor und bietet potenzielle Anwendungen zum Verständnis realer ökologischer Systeme.

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1 Introduction

In the perpetual dance of predator and prey within ecosystems, the strategies employed by both parties dictate the outcomes of life-and-death encounters. As predators evolve increasingly sophisticated hunting tactics, prey species must continually adapt to evade capture and ensure survival. One such strategy is the formation of cohesive groups, which has evolved across various species and scales, resulting in a wide range of collective behaviours. For instance, bird flocks often exhibit a beneficial hierarchical structure, optimizing their movement and resource use [1]. Similarly, ant colonies rely on chemical trails for communication and survival, showcasing the importance of group coordination [2]. Other examples include the synchronized movements of marching locusts [3], the collective decision-making processes in fish shoals [4] and the complex interactions observed in bacterial swarms [5]. Swarming behaviour provides various benefits. A notable advantage is the formation of a collective intelligence, enabling groups to make more effective decisions than individuals [6]. This collective approach allows for shared information processing, enhancing the utilization of individual sensory inputs [7].

Additionally, group formations can significantly increase prey survival chances during predator encounters. This is achieved through mechanisms such as information sharing to determine optimal escape routes [8], dilution of predation risk, heightened group vigilance, and the confusion of predators [9]. However, group aggregation is not without its downsides. Large groups can be more easily tracked and found by predators, presenting a significant risk [10]. Furthermore, the competition for limited resources among group members can lead to internal conflicts and stress. Despite the inherent challenges of group living, the benefits of collective behaviour in evading predators and enhancing survival remain a testament to the adaptive ingenuity of prey species in the wild. Examples of how animals escape from predators in groups are dwarf-herring that use flash-expansion, a swarm expansion in a spherical pattern, and the fountain effect, the splitting the group in two parts to turn and get behind the predator [11]. Another example are sand-eels, that most commonly use cruise and avoid, meaning a simple course correction of the group to avoid the predator in addition to herding, which means to stay a minimum distance away from the predator regardless of the position of the other group members [12]. In addition various species utilize protean escape strategies, evading the predator in a unpredictable manner by changing course, speed and movement patterns. This is done to make the prediction of the preys' movement very difficult.

The aim of this thesis is to use a torque based predation and group cohesion model to find optimal prey escape strategies through collective behavior. Patterns of collective motion, such as a predator hunting a group of prey, occur at various scales and in many different systems, which makes the development of simple statistical models viable to describe real phenomena seen in nature [13]. For example adding a simple noise term to the simulation can account for various environmental factors [13].

The understanding of complex collective motion can lead to benefits such as predictions of

the global positions of fish, insight into the collective reaction of people to panic or better comprehension of animal migration [13]. In this thesis an extended model of Active Brownian Particles (ABPs) is used. ABPs are self-propelled particles that undergo Brownian motion which refers to the random motion of particles suspended in a fluid due to collisions with the surrounding molecules [14]. Self-propulsion means that the particles can move autonomously, it can result from various sources such as chemical reactions, energy gradients, or motor proteins [15]. The interplay between self-propulsion and Brownian motion can lead to phenomena such as phase transitions, collective motion, and pattern formation in systems containing multiple ABPs [16]. That makes them a viable theoretical model to study collective motion such as a predator prey situation.

The extended ABP model introduces torque interactions between prey-prey and predator-prey particles. Prey particles utilize these torques to align and form cohesive groups, while simultaneously avoiding the predator. Conversely, predators experience torques that orient them towards prey, compelling them to move in pursuit. This torque-driven interaction, combined with active Brownian motion, forms a simple yet effective model for simulating predator-prey dynamics.

This thesis begins with an introduction to collective behavior, active Brownian particles (ABPs), and their torque interactions. Following this, the simulation methods and parameters are detailed, leading to a comprehensive analysis of the results.

2 Theory and Model

The following sections explore various patterns of collective motion and simulation models for predator-prey dynamics found in the literature. This is followed by an in-depth explanation of ABPs, encompassing the expanded ABP model utilized for simulations as well as the various interactions within it.

2.1 Different patterns of collective motion

The aim of this work is to reproduce predator-prey dynamics using a simple model, with prey particles engaging in different patterns of collective behavior. The following collective motion patterns are used to describe the group dynamics occurring in the simulations [13]:

- Disordered or cloud-like motion means all particles move in random directions.
- Fully-ordered motion describes all particles moving in the same direction.
- Rotational motion is a collective movement of all particles along a circle or rectangle.
- Critical behavior occurs when particle groups of different sizes move coherently in different directions. Critical refers to the fact, that the whole system is very sensitive to perturbations.
- Ripples refer to spatial and temporal patterns of local disturbances or fluctuations that propagate through the group.
- Quasi-long range velocity correlation describes a movement pattern where velocities of entities within a collective exhibit correlations over relatively large spatial scales.
- Jamming refers to a phenomenon where the movement of individuals or entities within a group becomes hindered or restricted, often leading to congestion or a halt in motion.

The occurrence of these patterns is system-independent and can be regarded as universal [13]. In literature various models were conceived that aim to reproduce these patterns as well as a predator prey-dynamic.

2.2 Other predator-prey models

To understand the complex behavioral patterns in predator-prey situations, different theoretical models were conceived. In this section, models from literature and their results are explained.
In source [17] the impact of the cooperative interaction range on the prey's survival chance is studied by using a simple force-based predator-prey model. The results show that short-range and long-range interactions are ineffective for the escape mechanism. However, an

intermediate range of interaction maximizes the survival probability of the prey group.
The model discussed in "Hunting ABP: Learning Optimal Behavior" [18] utilizes reinforcement learning to determine optimal predator behavior and interaction rules. Similar to this thesis, the paper also uses ABP as the basis of its model but uses a different interaction mechanism. The particles can react to environmental cues by executing a limited number of actions, such as adjusting their movement and turning either left or right. The key finding is that turning away from the predator is the strategy with the lowest predator success.
The paper in source [8] investigates chase-and-escape dynamics in group scenarios on a lattice, focusing on the effects of prey aggregation. In contrast to the model used in this work, multiple predators are employed. The paper demonstrates that aggregation significantly enhances the survival time of prey groups.
The study in source [19] employs a simplified individual-based model to explore collective predation dynamics through spatial animal movements. The model utilizes Vicsek-like rules, incorporating intra-group repulsion forces to simulate two groups of self-propelled organisms. The predators are attracted to, and prey are repelled from the nearest particles of the opposing group. The results show that increasing the number of predators decreases the total catch time, with an optimal escape strategy for the prey involving a weighted average of predator directions.
In this work, a model is used that relies on interacting Active Brownian particles.

2.3 Brownian Particles

In 1827, botanist Robert Brown observed the unpredictable movement of tiny pollen grain particles within a liquid [20]. Initially, he interpreted them as living organisms. However, Brown conducted further experiments with different materials, such as granular and glassy substances. Through this investigation, he determined the particles' purely physical nature. Despite being credited with identifying the phenomenon known as Brownian motion, Brown was not the first to observe it. Jan Ingen-Housz had already reported a similar irregular motion of coal dust particles immersed in a fluid in 1784 [21][22]. Brownian motion arises due to the thermal motion of the fluid molecules, which causes the particles to undergo continuous, random movements. This motion is characterized by its randomness and is described by statistical mechanics [21]. The equations of motion for overdamped Brownian particles are [14]:

$$\dot{\mathbf{r}}_i = \sqrt{2D}\boldsymbol{\xi}_i(t) \tag{1}$$

$$\dot{\Phi}_i(t) = \sqrt{2D_R}\eta_i(t) \tag{2}$$

with $\mathbf{r}_i$ being the particle i's location and Φ_i its orientation angle. The particle's motion is characterized by the translational and rotational diffusion coefficients $D = \mu k_B T$, $D_R = \mu_R k_B T$ respectively. They are specified by their respective mobilities μ and μ_R, which are measures of the particle's tendency to move due to thermal fluctuations, with T being the

temperature and k_B the Boltzmann constant. The random movement of the particles is caused by the two Gaussian white noise distributions $\boldsymbol{\xi}_i(t)$ and $\eta_i(t)$, that are characterized by having zero mean, constant variance, and being uncorrelated in time, thus modeling the unpredictable and random nature of thermal fluctuations. In the non-active case, the equations for orientation Φ_i and location $\mathbf{r}_i$ are decoupled [14].

2.4 Active Brownian Particles

In contrast to Brownian particles, which move solely due to thermal fluctuations, Active Brownian Particles are propelled by an internal energy source, such as chemical reactions or external energy fields [21]. The equations in the active case are [14] the following:

$$\dot{\mathbf{r}}_i = v_0 \mathbf{u}_i + \sqrt{2D}\boldsymbol{\xi}_i(t) \tag{3}$$

$$\dot{\Phi}_i(t) = \sqrt{2D_R}\eta_i(t) \tag{4}$$

with v_0 being the propulsion velocity and $\mathbf{u}_i = (\cos\Phi_i, \sin\Phi_i)^{\mathrm{T}}$ the particle's orientation. $\boldsymbol{r}_i$, Φ_i, D, D_R, $\boldsymbol{\xi}_i$ and η_i are as defined in section 2.3. Systems of interacting ABPs can be used to model swarm motion, meaning a confined system of particles moving collectively [21]. Assuming patterns of collective motion are universal [13], it is possible to model real-world swarms as interacting ABPs[23].

2.5 The ABP model used

The model of interacting ABPs used in this thesis is based on source [24].
In this paper, the collective behavior of a two-dimensional suspension of self-propelled particles is examined. These particles move at low Reynolds numbers and interact through non-reciprocal pairwise interactions [24]. A low Reynolds number indicates a flow regime where viscous forces dominate over inertial forces [24]. The particles undergo active Brownian motion, with the equations of motion found in equation 4. The interactions between the particles are twofold: Firstly, particles experience repulsive steric forces within a typical interaction range, which can be seen as the effective diameter of the particles. Additionally, the particles are subject to pairwise orientational interactions over a short range. These interactions manifest as torques. Importantly, these orientational interactions are non-reciprocal, meaning the torque on particle i due to particle j, τ_{ij}, is not simply the negative of the torque on particle j due to particle i τ_{ji}. Instead, it depends on the orientation of particle i and the separation vector between particle i and j. These interactions cause particles to turn away from each other.
The model in source [24] is expanded upon to reproduce a predator-prey dynamic. The expanded model aims to be a representation of the complex dynamics of predator-prey behaviors. The prey particles are moving in a group and experience cohesive and aligning torque interactions, while staying a minimum distance away from each other. At the same

time, they try to turn and get away from the predator, who hunts after them. When a prey comes within a certain distance, or "kill radius" of the predator r_{E}, it gets eaten.
A system comprising $N = 100$ actively moving prey particles and one predator particle in two spatial dimensions, where each particle moves with a constant speed v_0 and can change its orientation $\mathbf{u}$ is examined. The change of orientation is governed by the particles' interaction torques $\tau_{i,j}$ described in the sections 2.5.2 and 2.5.3. In addition to that, particles experience repulsive steric forces $\boldsymbol{F}_{i,j}$ at a typical interaction range resulting from the WCA potential [25] described in section 2.5.1. The equations of motion are:

$$\dot{\mathbf{r}}_i = v_0 \mathbf{u}_i + \mu \sum_{i \neq j} \mathbf{F}_{i,j} + \sqrt{2D} \boldsymbol{\xi}_i(t) \tag{5}$$

$$\dot{\Phi}_i(t) = \mu_R \sum_{i \neq j} \tau_{i,j} + \sqrt{2D_R} \eta_i(t) \tag{6}$$

where D, D_R, μ, μ_R, η_i and $\boldsymbol{\xi}_i$ are as defined in sections 2.3 and 2.3. This is a system of three coupled differential equations.

2.5.1 WCA Potential

The WCA (Weeks-Chandler-Andersen) potential is a modified form of the Lennard-Jones potential. It is designed to model purely repulsive interactions, removing the attractive part of the Lennard-Jones potential. Effectively, the WCA potential prevents particles from overlapping. The strong repulsive interactions start to take effect at the cutoff distance $r_c = 2^{\frac{1}{6}}\sigma$ and have the form:

$$U_{\epsilon,\sigma} = \begin{cases} 4\epsilon \left(\left(\frac{\sigma}{|\boldsymbol{r}|} \right)^{12} - \left(\frac{\sigma}{|\boldsymbol{r}|} \right)^{6} \right) + \epsilon & |\boldsymbol{r}| \leq r_c \\ 0 & |\boldsymbol{r}| > r_c \end{cases} \tag{7}$$

with $\epsilon / k_B T = 100$ being the energy scale [25]. The forces caused by the WCA potential take the form $\boldsymbol{F}_{i,j} = \boldsymbol{\nabla}_{r_i} U_{\epsilon,\sigma}(r_i - r_j)$. The WCA potential is displayed graphically in figure 1. In addition to the WCA potential, the particles interact via non-reciprocal torques that represent prey-prey and predator-prey interactions [24].

2.5.2 Prey-prey interactions

The prey-prey interactions are split into alignment and cohesion interactions that take effect within a prey-prey particle interaction range of R_{pp}. The alignment and cohesion interactions are shown in figure 2a and 2b and are defined by:

$$\tau_{i,j} = T_{\mathrm{A}} \theta(R_{\mathrm{pp}} - |\boldsymbol{r}_{i,j}|) \sin(\Phi_j - \Phi_i) \tag{8}$$

$$\tau_{i,j} = -T_{\mathrm{AA}} \theta(R_{\mathrm{pp}} - |\boldsymbol{r}_{i,j}|) \left| \boldsymbol{u}_i \times \frac{\boldsymbol{r}_{i,j}}{|\boldsymbol{r}_{i,j}|} \right| . \tag{9}$$

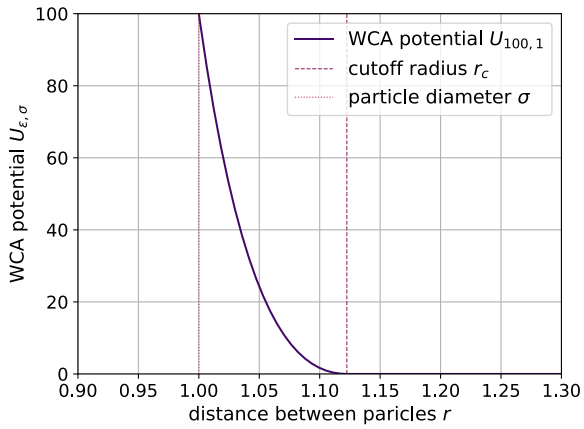


Figure 1: **WCA potential:** The figure shows the WCA potential $U_{\epsilon,\sigma}$ for $\epsilon = 100$ and $\sigma = 1$ over the particle distance r. The cutoff radius r_c and the particle diameter σ are also shown.

respectively, with the alignment and cohesion torque amplitudes T_{A} and T_{AA}, the heavyside step function θ, the distance between the particles i and j $\boldsymbol{r}_{i,j}$ and the particles' orientation angle Φ_i and vector $\boldsymbol{u}_i$. The relation between the torque amplitudes is $T_{\mathrm{AA}} = \frac{T_{\mathrm{A}}}{2}$. Hence, the total torque resulting from both the cohesion and alignment interactions takes the following form:

$$\tau_{i,j} = T_{\mathrm{A}}\theta(R_{\mathrm{pp}} - |\boldsymbol{r}_{i,j}|)\left(\sin\left(\Phi_j - \Phi_i\right) - \frac{1}{2}\left|\boldsymbol{u}_i \times \frac{r_{i,j}}{|r_{i,j}|}\right|\right) \tag{10}$$

2.5.3 Predator-prey and prey-predator interactions

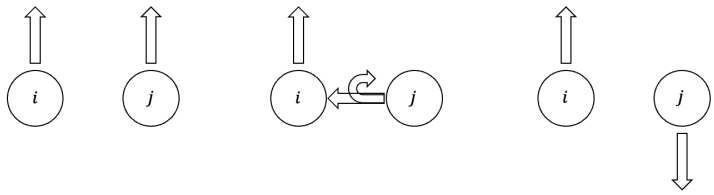
When the prey particle j enters the interaction range of the predator R_{P} it exerts a torque upon the predator $\tau_{\mathrm{Pred},j}$ that takes the form:

$$\tau_{\mathrm{Pred},j} = -T_0\theta(R_{\mathrm{P}} - |\boldsymbol{r}_{\mathrm{Pred},j}|)\left|\boldsymbol{u}_j \times \frac{\boldsymbol{r}_{\mathrm{Pred},j}}{|\boldsymbol{r}_{\mathrm{Pred},j}|}\right|. \tag{11}$$

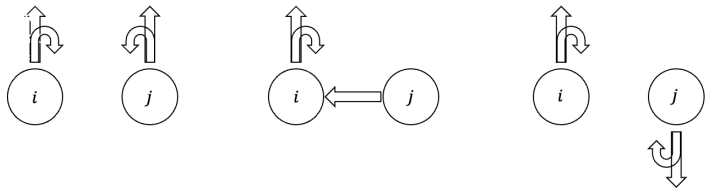
with the Heavyside step function θ and the predator-prey and prey-predator interaction torque amplitude T_0. The distance between the predator and the prey particle j is described by $|\boldsymbol{r}_{\mathrm{Pred},j}|$, the orientation vector of particle j by $\boldsymbol{u}_j$. In contrast, the prey particles experience the torque $\tau_{j,\mathrm{Pred}}$ when the predator enters their interaction radius R_{p}:

$$\tau_{j,\mathrm{Pred}} = T_0\theta(R_{\mathrm{p}} - |\boldsymbol{r}_{j,\mathrm{Pred}}|)\left|\boldsymbol{u}_j \times \frac{\boldsymbol{r}_{j,\mathrm{Pred}}}{|\boldsymbol{r}_{j,\mathrm{Pred}}|}\right|. \tag{12}$$

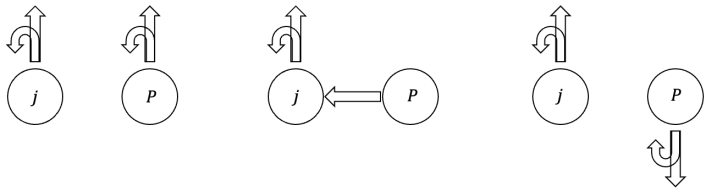
with the heavyside step function θ and the distance between the particle j and the predator $\boldsymbol{r}_{j,\mathrm{Pred}}$. Both predator-prey and prey-predator interactions are shown in figure 2c.



(a) Alignment interaction between prey particles i and j.



(b) Cohesion interactions between prey particles i and j.



(c) Interactions between predator P and prey j

Figure 2: Torque-based interactions between particles

3 Methods

In this section, the numerical methods used for this simulation, the simulation duration, parameters and duration, as well as the methods used for the data analysis, are discussed.

3.1 Numerical Simulation

In this study a computational approach is employed to simulate the dynamics of interacting active Brownian particles (ABPs) in a two-dimensional environment. The simulation was implemented using the Python programming language, with the NumPy library for numerical computations and Matplotlib for visualization.

3.1.1 Euler Method

To simulate the trajectories of the interacting ABPs, the Langevin equations are discretized using the Euler method [26]. The Euler method is a numerical technique used to find solutions to ordinary differential equations. It's one of the simplest methods for solving these equations numerically. The method is based on the idea of approximating the solution curve by successively moving forward small steps along the tangent line at each point. The method starts with an initial condition, meaning the value of the function and its derivative at a certain point in time. Next, a small step size h is chosen, which determines how far each step moves along the tangent line. This process is iterated through the differential equation, updating the value of the function at each step using the slope of the tangent line at the current point. When $y(t_n)$ is the approximate value of the solution at time t_n, the Euler method looks like:

$$y_{n+1} = y_n + h \cdot f(t_n, y_n) \tag{13}$$

with $f(t, y)$ being the function defining the differential equation. So $f(t_n, y_n)$ is the slope at the tangent line at $y(t_n)$ at time t_n. By multiplying it with h and adding it to the current value of y_n, you approximate the next value, y_{n+1}. This process is repeated until the desired amount of timesteps is reached. The position and orientation of each particle are updated iteratively at discrete time steps.

3.1.2 Application

To discretize the equations of motion (5) from section 6, the simulation time is divided into small timesteps with the length $\Delta t = 10^{-5}$. At each time step $t_n = n \cdot \Delta t$, the position and

orientation of particle i: $\boldsymbol{r}_i(t_n)$ and $\Phi_i(t_n)$ are updated along the following scheme:

$$\mathbf{r}_{i,n+1} = \boldsymbol{r}_{i,n} + \Delta t(v_0\mathbf{u}_{i,n} + \mu \sum_{i\neq j} \mathbf{F}_{i,j,n}) + \sqrt{2D\Delta t}\boldsymbol{\omega}_{r,i}(t) \tag{14}$$

$$\Phi_{i,n+1} = \Phi_{i,n} + \Delta t\mu_R \sum_{i\neq j} \tau_{i,j,n} + \sqrt{2D_R\Delta t}\omega_{\Phi,i}(t) \tag{15}$$

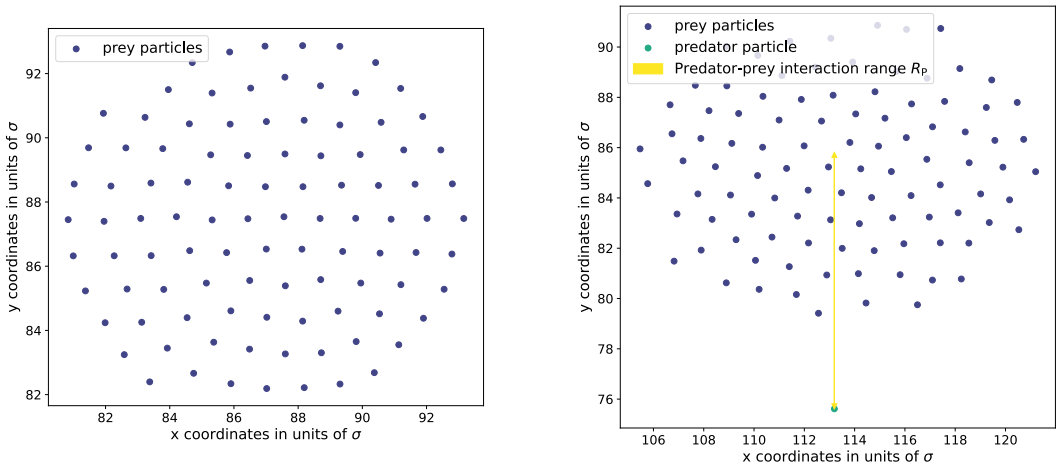
The position of particle i at the time step $n+1$: $\boldsymbol{r}_{i,n+1}(t_{n+1})$ is calculated from its old position at n: $\boldsymbol{r}_{i,n}$ and the particle's movement resulting from its speed and the WCA-potential multiplied by the length of the time step Δt. The random motion of the particle is facilitated by the Wiener process $\boldsymbol{\omega}_{r,i}(t)$. The particle's orientation angle $\Phi_{i,n+1}$ is calculated analogously. This represents a numerical solution of the equations of motion.

3.1.3 Simulation parameters

The particle diameter and translational mobility are set to $\sigma = 1$ and $\mu = \frac{1}{3}$. That means that the rotational mobility equals $\mu_R = 1$. With k_BT also set to $k_BT = 1$, the translational and rotational diffusion coefficients are $D = \frac{1}{3}$ and $D_R = 1$. The energy scale of the WCA potential interaction is $\epsilon = 100$ and the cutoff radius is $r_\mathrm{c} = 2^{\frac{1}{6}}\sigma = 2^{\frac{1}{6}}$. The simulation area is $A = 175\,\sigma \cdot 175\,\sigma$ and has periodic boundary conditions. It can be characterized by the packing fraction $\theta = N\sigma^2\pi/(4 * A) = \pi/1225 \approx 0.0026$ and the Peclet number $\mathrm{Pe} = \sigma v_0/D = 26\frac{2}{3}$, with the particle speed being $v_0 = 26\frac{2}{3}$ for number of particles $N = 100$. The predator's speed is set to $v_\mathrm{P} = 40$. Both the strength of the torque interaction T_A and T_0, as well as the interaction radii R_pp, R_p from the torque prey-prey and prey-predator interactions (equations (9) and (12)) are varied in the simulations. The "kill radius" r_E of the predator, which is the distance within which it captures the prey, is set to $2r_\mathrm{c}$. The simulation time is measured in units of τ_r with $\tau_\mathrm{r} = \frac{1}{D_R}$.

3.1.4 Simulation setup

The simulation is set up to achieve results that are as comparable as possible. The initial setup is displayed in figure 3. In every simulation, the prey particles are arranged in concentric circles around the center of the simulation area (figure 3a), with their orientations pointing inwards. The prey particles are given a time of $t = 1\,\tau_\mathrm{r}$ to move and leave the setup position. After this period, the predator spawns one interaction range R_P below the center of the prey particles, with its orientation vector pointing upwards (figure 3b). The result is an initial phase where the predator interacts with the prey group and a second phase where the predator moves around hunting for the prey.



(a) **Simulation setup:** The prey particles are arranged in concentric circles around the center of the simulation.

(b) **Simulation after $t = 1\ \tau_{\mathbf{R}}$:** The predator spawns one interaction range R_{P} below the center of the prey group

Figure 3: Simulation setup

3.1.5 Simulation duration

The duration of the simulations is set to 30 τ_{r}. The choice of this comparatively short simulation duration has two main reasons:

- The (small) number of simulated particles $N = 100$, leads to a majority of the particles (around 50-80%) getting eaten before the simulation is over. That means, that the predator-prey dynamic is apparent in all simulations, despite their (short) duration.
- A shorter simulation duration allows for a large parameter scan. This is very important to get a good picture of the dynamics, since there are many parameters that can be varied.

Each simulation is run three times, resulting in three trials.

3.2 Simulation data analysis

To analyze the results of the simulations, multiple methods are used. The proximity coefficient (section 3.2.1) and the average particle distance (section 3.2.2) determine how close the prey particles stick together and provide a quantitative measurement for their group structure. The survivorship function (section 3.2.3) shows the number of prey still living.

3.2.1 Proximity coefficient

The proximity coefficient P is a metric for determining the average number of particles within the interaction radius R_{pp} of an average prey particle per time step. It is calculated by counting the (alive) particles within the radius R_{pp} of each particle for each time step and then dividing by the total time steps T and particles N:

$$P = \frac{1}{TN} \sum_{t=1}^{T} \sum_{i=1}^{N} \left(\sum_{\substack{j=1 \\ j \neq i}}^{N} \mathbb{I}(d_{ij}(t) \leq R_{\mathrm{pp}}) \right) \tag{16}$$

with the time step $t \in \{1, ..., T\}$ and the indicator function $\mathbb{I}$, that equals 1 if the distance between particles i and j $d_{ij}(t)$ is less than or equal to R_{pp}.

3.2.2 Average particle distance

The average distance of the prey particles D is a measurement of how close they are together and what the group structure looks like. It is calculated by averaging the distance between all surviving prey particles per time step t and particle i.

$$D = \frac{1}{TN} \sum_{t=1}^{T} \sum_{i=1}^{N} \left(\sum_{\substack{j=1 \\ j \neq i}}^{N} d_{ij}(t) \right) \tag{17}$$

with the time step $t \in \{1, ..., T\}$ and the distance between particles i and j: $d_{ij}(t)$. The higher the distance between the particles, the smaller the proximity coefficient P.

3.2.3 Survivorship function

The survivorship function $S(t)$

$$S(t) = N - \sum_{t > t_{\mathrm{E}}} E(t_{\mathrm{E}}) \tag{18}$$

measures how many prey are alive at a certain point in time t. The number of eating events $E(t_{\mathrm{E}})$ until a certain point in time t_{E} is substracted by the number of prey at the start $N = 100$.

4 Results and Discussion

In this thesis, several simulations for different prey-predator interaction radii R_p and prey-predator interaction torques T_0 are studied. Each batch of simulations consists of 24 simulations, for each of the six different torques prey-prey interaction torque strengths $T_\mathrm{A} \in \{0.1, 1, 10, 20, 30, 40\}$, at each of the four different prey-prey interaction radii $R_\mathrm{pp} \in \{2r_\mathrm{c}, 3r_\mathrm{c}, 4r_\mathrm{c}, 5r_\mathrm{c}\}$. To understand the interactions between predator and prey, the prey-prey interactions without a predator are analyzed first.

4.1 Prey-prey interaction analysis

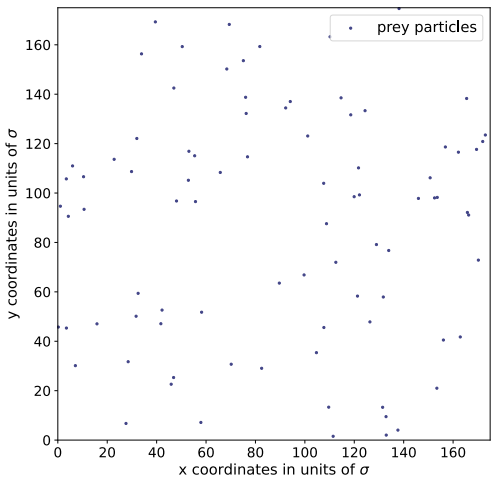
For this analysis a batch of simulations is run without a predator. For each simulation, three trials are carried out. The prey particles start on a grid so that they can more easily reach an equilibrium position. Each torque T_A interaction radius R_pp combination is allocated to one of four prey behavior states, these are shown in figure 4. A state diagram is displayed in figure 5. The different prey state are:

- Cloud: The particles move in random directions and their interactions with each other are negligible (figure 4a). This behavior is referred to as disordered or cloud-like in section 2.1.
- Multiple loose groups: Some particles move in groups of different sizes, others by themselves. The groups dissolve and emerge over time (figure 4b). This can be seen as a mixture of ordered and cloud like behavior (section 2.1).
- Multiple cohesive groups: The particles are split into cohesive groups that generally stay intact (figure 4c). This movement pattern is called critical, because the system is very sensitive to perturbations [13] (section 2.1).
- One group: All particles stick together in one group and cohesively move in one direction (figure 4d). This behavior corresponds to fully-ordered movement from section 2.1

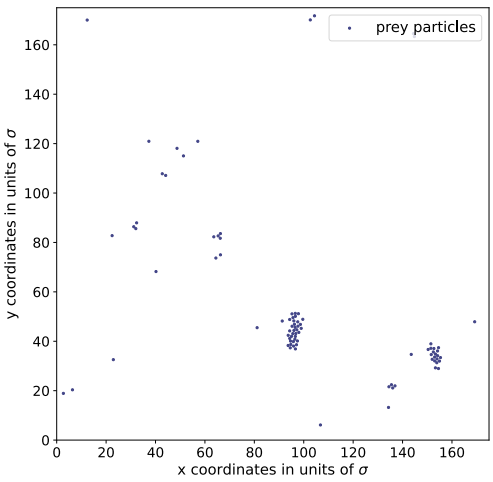
With this analysis done, a predator is introduced into the system.

4.2 Predator-prey interaction analysis at different R_p

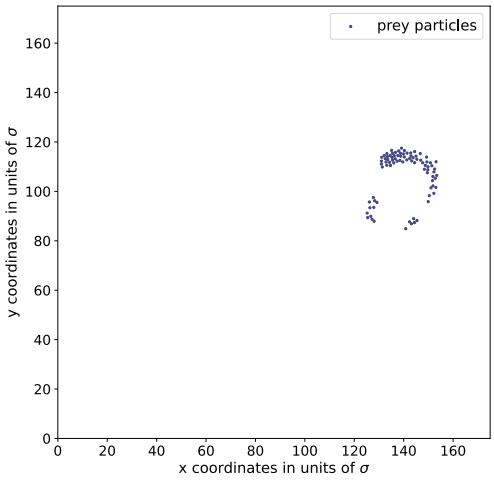
In this section the simulations of the predator-prey system are examined. A batch of simulation is carried out analogously to section 4.1 by iterating through torques T_A and interaction radii R_pp. In each simulation a different torque T_0 and interaction radius R_p between prey and predator are used and the results are compared. Due to the fact that one prey particle interacts with multiple other prey particles, but only with one predator, the torque interaction strength T_0 between prey and predator is set to double the prey-prey torque



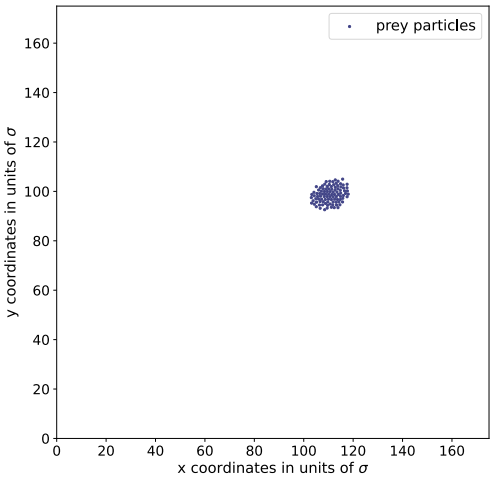
(a) Prey particles in a cloud-like state.



(b) Prey particles in loose groups.



(c) Prey particles in multiple cohesive groups.



(d) Prey particles in one consistent group.

Figure 4: Different states of the prey particles.

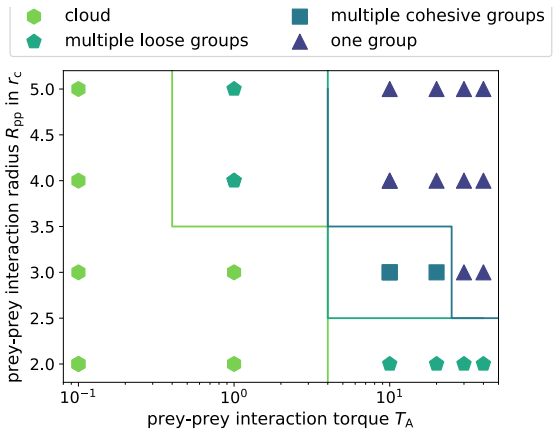


Figure 5: **State diagram:** Prey particles' state as a function of torque T_A and prey-prey interaction radius R_{pp}.

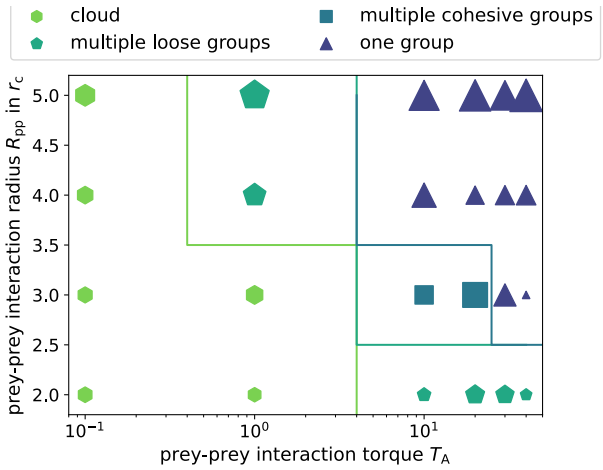
interaction strength $T_0 = 2T_A$. Simulations are done for prey-predator interaction radii of $R_p \in \{3r_c, 5r_c, 9r_c\}$. The setup parameters for these simulations are detailed in Section 3.1.3, while the results are discussed in the following sections.

4.2.1 Dynamics at small interaction radii

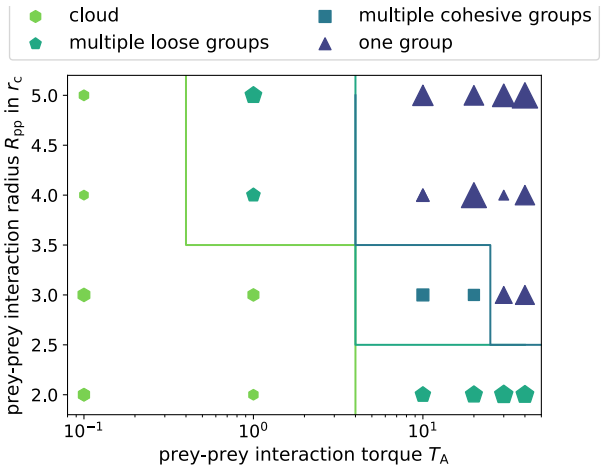
In this simulation batch, a prey-predator interaction radius of $R_p = 3r_c$ and a torque of $T_0 = 2T_A$ are used. The results are displayed in figures 6a, 7 and 8. In figure 7 the survivors over time for all simulations are shown. At the beginning of all simulations in the batch, there is an accelerated decrease in the number of surviving prey. This behavior is due to the initial simulation setup (section 3.1.4), where the predator starts oriented towards and within range of a larger concentration of prey, allowing it to catch many prey particles early on.

A low predator success, characterized by an almost linear decrease in the number of survivors, is observed at low torque strengths of $T_A \in \{0.1, 1\}$ (also visible in figure 6a) for all prey-prey interaction radii R_{pp}, where prey movement is cloud-like. Due to random prey motion, the predator is almost always in range of prey and hunts them one by one, continuing this process until either no prey particles are left or the simulation ends. This results in rather low predator success, as the prey can avoid the predator effectively when their movement is not influenced by other prey particles. Consequently, predator success is low across almost all torques for the interaction radius $R_{pp} = 2r_c$. The number of survivors varies significantly depending on the prey-prey interaction radius R_{pp}, with the lowest number of survivors occurring at $R_{pp} = 5r_c$.

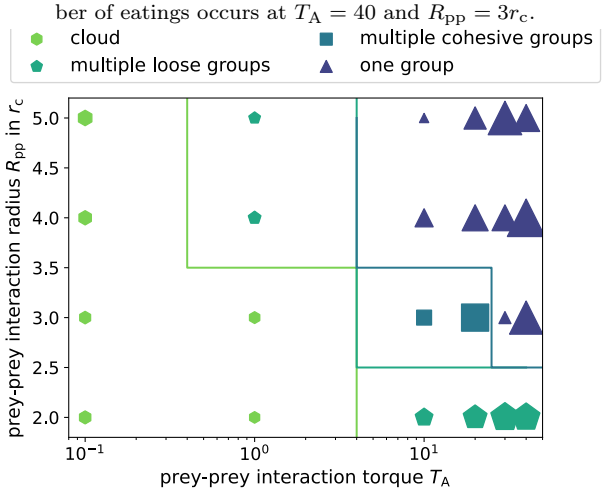
For higher torques $T_A \geq 10$, the graphs exhibit a step-like pattern. This occurs because cohesive prey movement leads to the predator moving randomly for extended periods without seeing prey, as there is not always a prey particle within its interaction range. When the



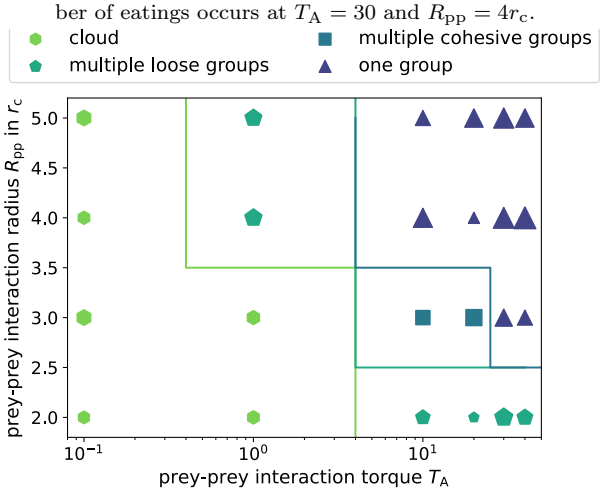
(a) **number of eatings for $R_p = 3r_c$:** The lowest number of eatings occurs at $T_A = 40$ and $R_{pp} = 3r_c$.



(b) **number of eatings for $R_p = 5r_c$:** The lowest number of eatings occurs at $T_A = 30$ and $R_{pp} = 4r_c$.



(c) **number of eatings for $R_p = 9r_c$:** The lowest number of eatings occurs at $T_A = 10, R_{pp} = 5r_c$ followed by $T_A = 30, R_{pp} = 3r_c$



(d) **number of eatings for $R_p = 3r_c$ and $T_0 = 2T_A$:** The lowest number of eatings occurs at $T_A = 20$ and $R_{pp} = 4r_c$.

Figure 6: **Number of eatings for all parameter combinations:** There is a plot for each of the simulation batches. Each marker in the plots represents one simulation, averaged over three trials. The marker shape depicts the prey's movement pattern at the specific prey-prey interaction parameters and the marker-size represents the number of eaten prey.

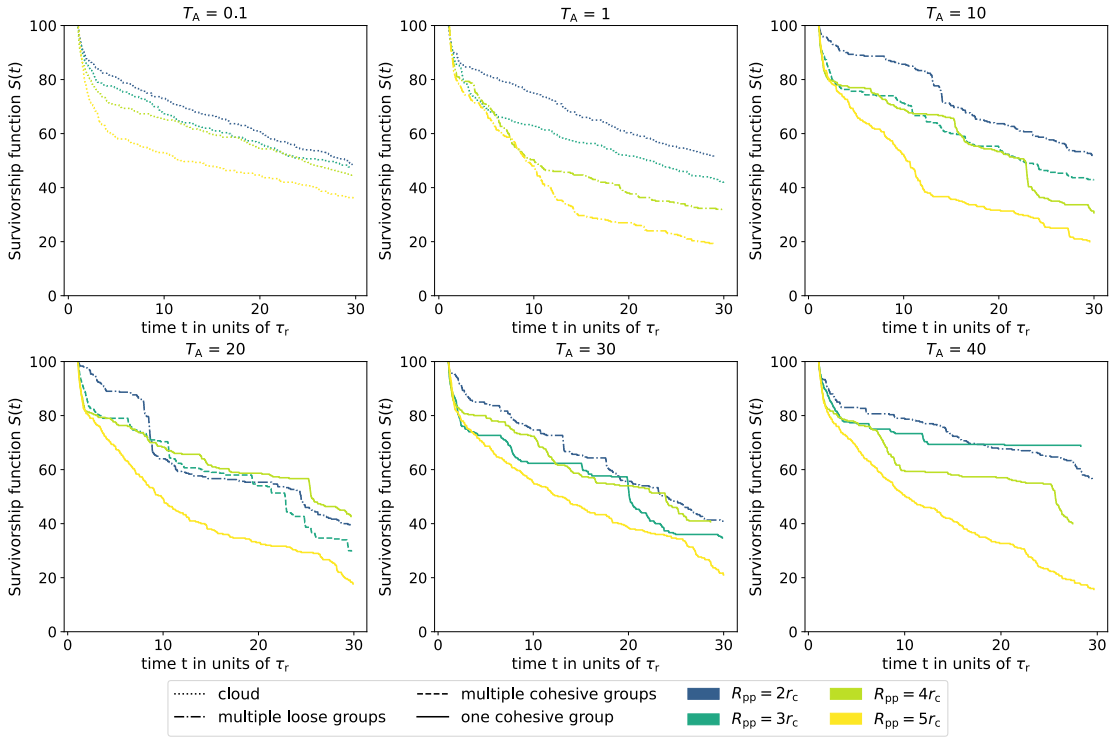


Figure 7: **Survivors over time at $R_p = 3r_c$:** This plot shows the number of survivors over time for all 24 simulations for the prey-predator interaction radius of $R_p = 3r_c$, each averaged over three trials. The linestyle shows the prey state explained in section 2.1. A solid line represents the prey moving as one cohesive group; the dashed line means there are multiple cohesive groups; the dashed-dotted line refers to the prey moving in multiple loose groups; and the dotted line indicates there is a cloud-like prey movement. The colors represent the different prey-prey interaction radii R_{pp}.

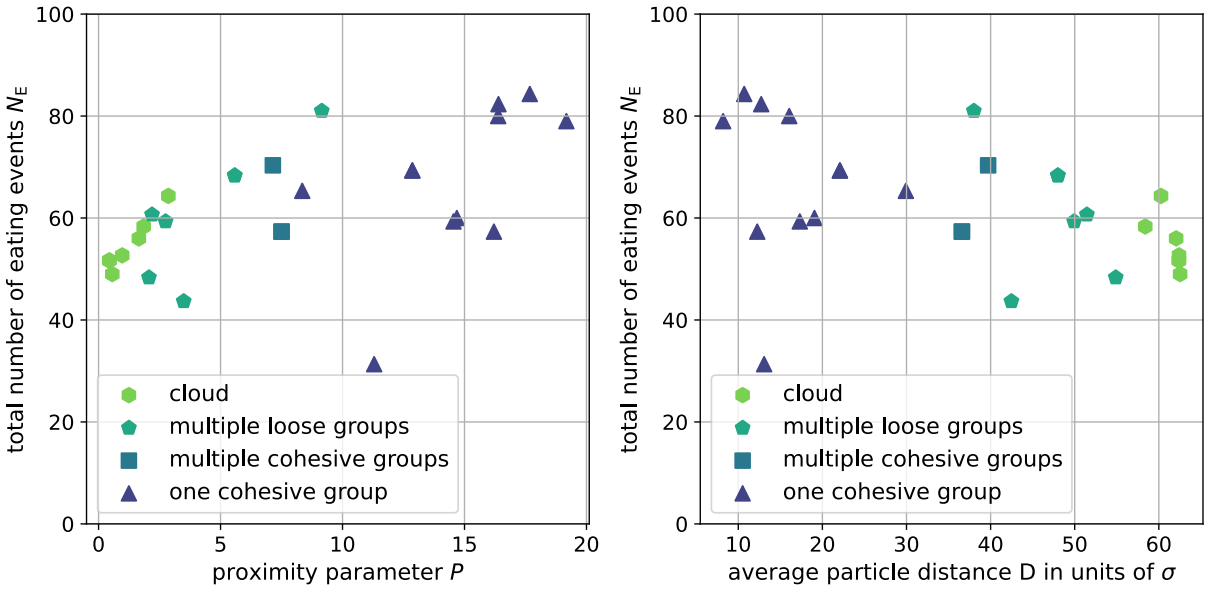


Figure 8: **Number of eating events for prey proximity and distance at $R_p = 3r_c$:** This graph shows the correlation between the prey swarms' movement pattern and the number of eaten prey for the prey-predator interaction radius of $R_p = 3r_c$. In the left plot the proximity parameter (section 3.2.1) and in the right plot the distance parameter (section 3.2.2) are used.

predator eventually finds a group of prey, it consumes many in a short time. Generally, higher torques and interaction radii, which result in more cohesive prey movement patterns, yield high predator success, which is visible in figure 6a as well. This outcome arises from the complex interplay between the prey and predator and the interference of cohesion and avoidance mechanics. However, the parameter combination $T_A = 40$ and $R_{pp} = 3r_c$ results in a very low predator success across all three trials, despite the high torque and interaction radius.

In figure 8 the number of eating events is plotted over the proximity parameter and average prey-prey distance. The plot also shows a slight trend, that the predator success is higher for more cohesive groups. Predator success is smaller for more loose prey movement patterns. There are a few exceptions where the cohesive group's survival chance is equal to, and one exception where it is even greater than the survival chance of the loose prey group. This specific combination of interaction radius and torque yields the lowest predator success.

4.2.2 Dynamics at medium interaction radii

In this simulation batch a prey-predator interaction radius of $R_p = 5r_c$ and a torque of $T_0 = 2T_A$ are used. The results are displayed in figures 6b, 9 and 10. In figure 9 the survivors over time for all simulations are shown. In simulations with torque $T_A \geq 30$, predator

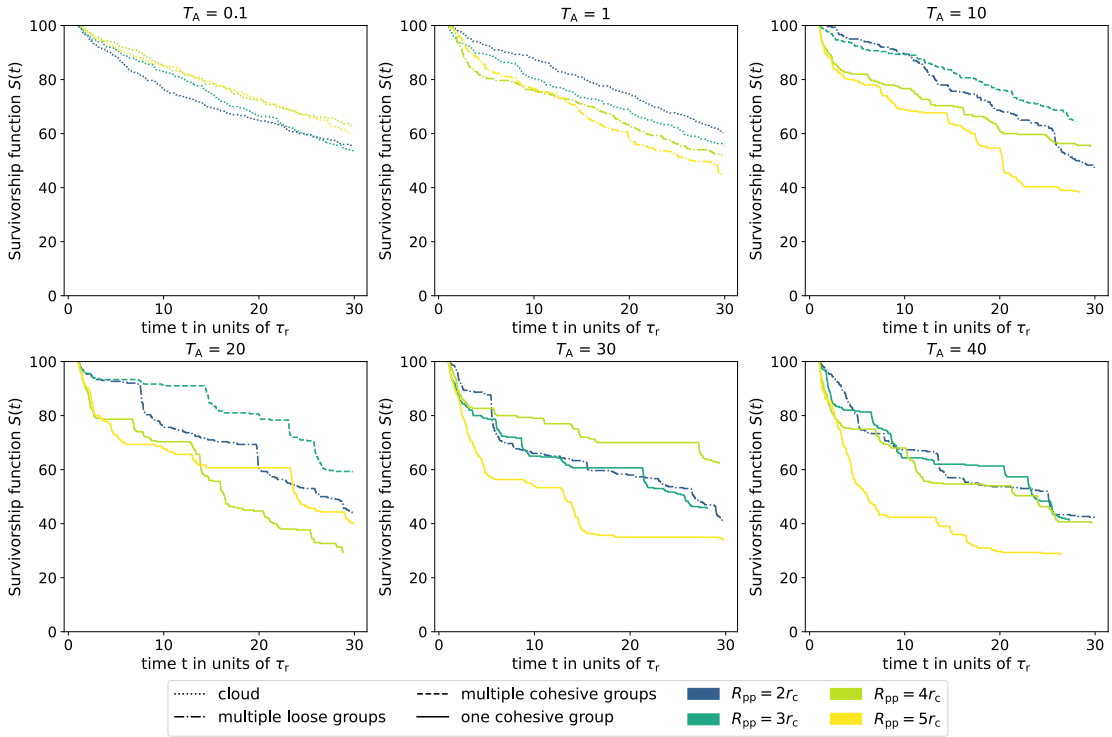


Figure 9: **Survivors over time at $R_p = 5r_c$:** This plot shows the number of survivors over time for all 24 simulations for the prey-predator interaction radius of $R_p = 5r_c$, each averaged over three trials. The linestyle shows the prey state explained in section 2.1. A solid line represents the prey moving as one cohesive group; the dashed line means there are multiple cohesive groups; the dashed-dotted line refers to the prey moving in multiple loose groups; and the dotted line indicates there is a cloud-like prey movement. The colors represent the different prey-prey interaction radii R_{pp}.

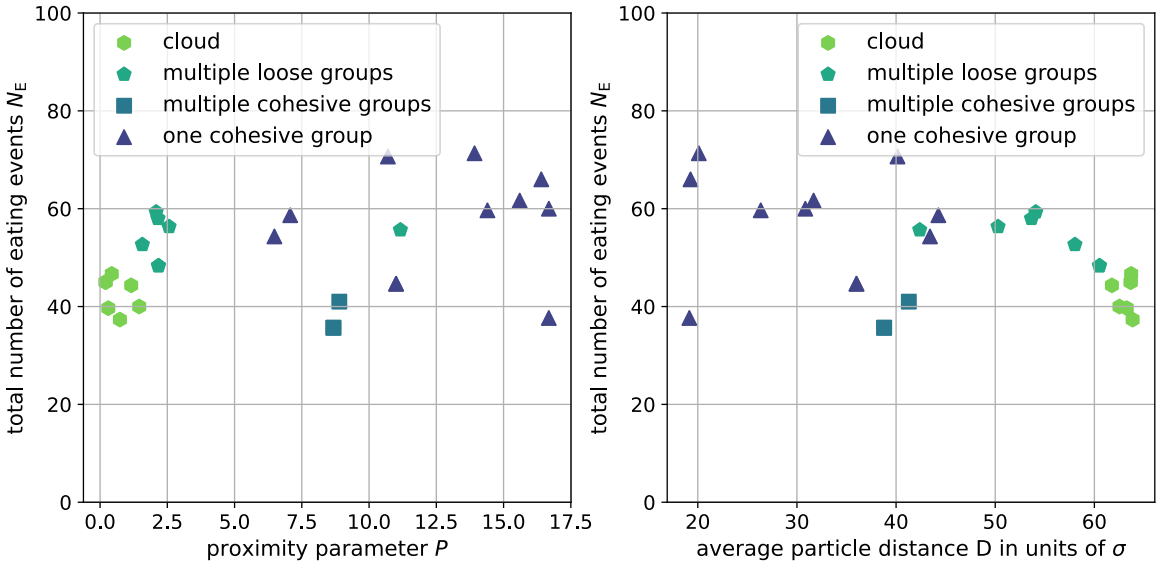


Figure 10: **Number of eating events for prey proximity and distance at $R_p = 5r_c$:** This graph shows the correlation between the prey swarms' movement pattern and the number of eaten prey for the prey-predator interaction radius of $R_p = 5r_c$. In the left plot the proximity parameter (section 3.2.1) and in the right plot the distance parameter (section 3.2.2) are used.

success is very high at the beginning, similar to the behavior described in section 4.2.2, but this trend is only visible at larger torques. This occurs because, at low torques, prey can detect and avoid the predator earlier, whereas cohesive groups at higher torques react slower due to stronger prey-prey interactions, interfering with predator evasion.

Overall, predator success at high prey-prey interaction torques is much higher than at low prey-prey interaction torques (also visible in figure 6b), as noted in section 4.2.2. An exception is the combination of interaction radius $R_{pp} = 4r_c$ and torque $T_A = 30$, where predator success reaches a maximum.

Similar to the dynamics at small prey-predator interaction ranges (section 4.2.2) R_p, the survivorship function shows an almost linear decline with low predator success at low torques of $T_A \in \{0.1, 1\}$. At larger torques ($T_A \geq 10$), the survivorship function has a step-like pattern, and predator success is higher. This results from predator evasion being more efficient in cloud-like prey movement states, as explained in section 4.2.2. At higher torques, the prey-prey interaction radius and predator success do not seem to be correlated. In contrast to section 4.2.2, the differences between prey-prey interaction radii for each torque are small, especially for lower torques.

In figure 10 the number of eating events is plotted over the proximity parameter and average prey-prey distance. It shows that a low proximity parameter P and a high average particle distance D (more loose prey movement states) correlate with a low predator success, with

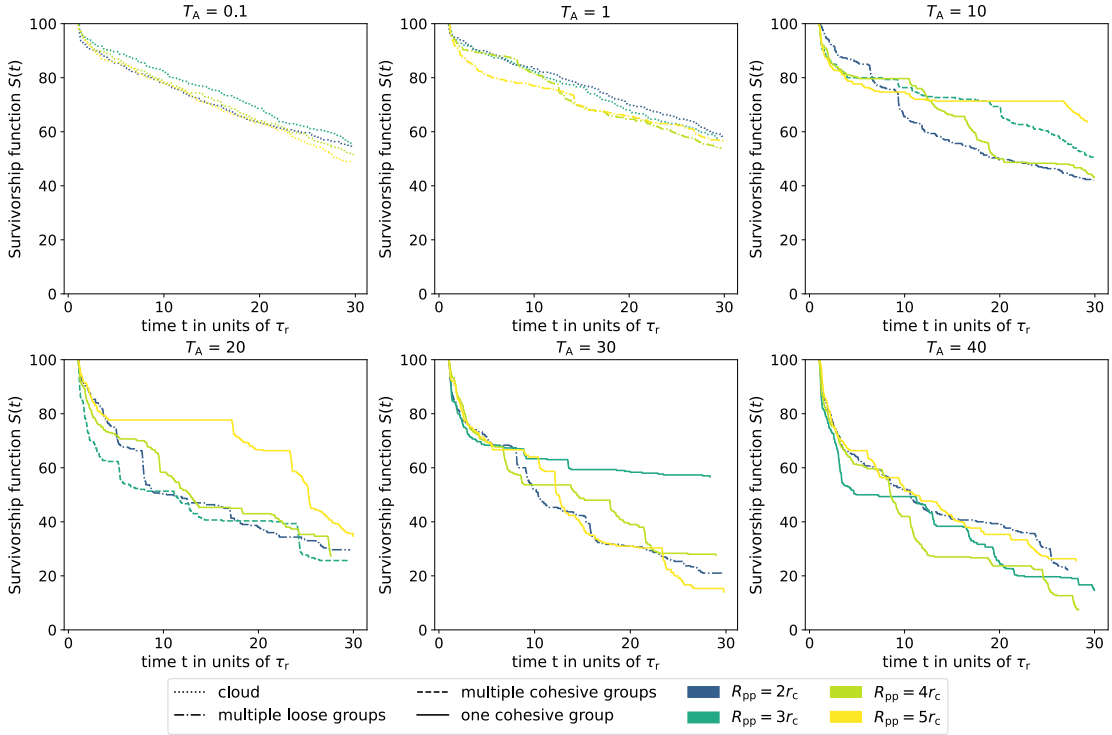


Figure 11: **Survivors over time at $R_p = 9r_c$:** This plot shows the number of survivors over time for all 24 simulations for the prey-predator interaction radius of $R_p = 9r_c$, each averaged over three trials. The linestyle shows the prey state explained in section 2.1. A solid line represents the prey moving as one cohesive group; the dashed line means there are multiple cohesive groups; the dashed-dotted line refers to the prey moving in multiple loose groups; and the dotted line indicates there is a cloud-like prey movement. The colors represent the different prey-prey interaction radii R_{pp}.

the one exception mentioned earlier. Conversely, a cohesive group movement pattern almost always results in high predator success, which is also visible in figure 6b. Noticeably, the two simulations where the prey move in multiple cohesive groups also yield low predator success.

4.2.3 Dynamics at large interaction radii

In this simulation batch a prey-predator interaction radius of $R_p = 9r_c$ and a torque of $T_0 = 2T_A$ are used. The results are displayed in figures 6c, 11 and 12. In figure 11 the survivors over time for all simulations are shown. For torques of $T_A \geq 10$, eating rates at the beginning of the simulations, shortly after the predator introduction, are very low, similar to the previous sections. There is an almost linear decline in the survivorship function at low torques of $T_A \in \{0.1, 1\}$, similar to other prey-predator interaction radii. At these torques, all graphs for the different prey-prey interaction radii R_{pp} are almost congruent. Overall, the graphs for each prey-prey interaction radius are closer together compared to other simulation

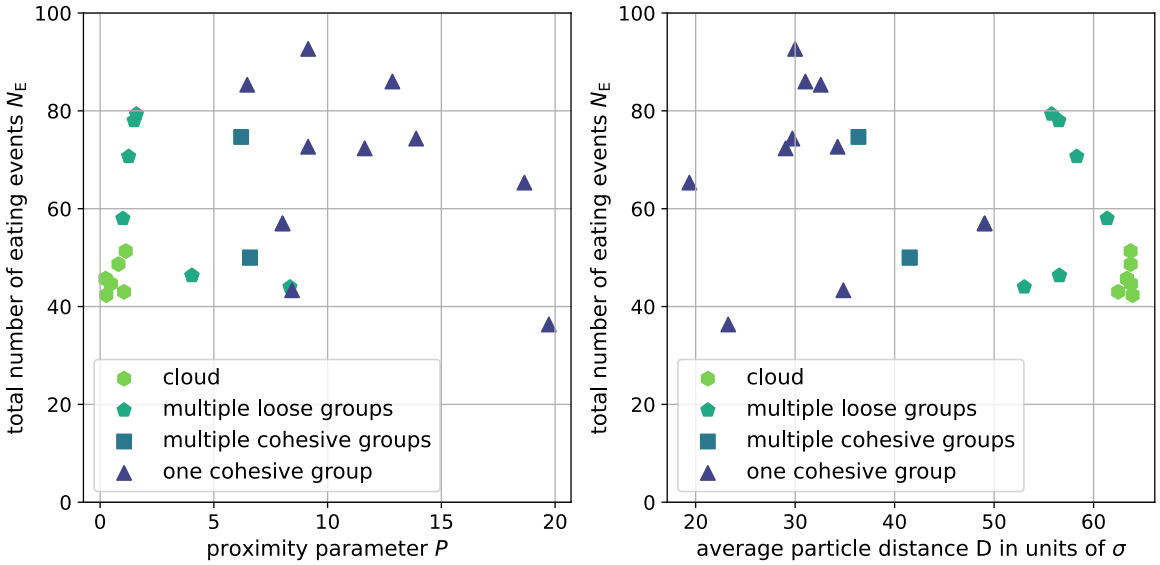


Figure 12: **Number of eating events for prey proximity and distance at $R_p = 9r_c$:** This graph shows the correlation between the prey swarms' movement pattern and the number of eaten prey for the prey-predator interaction radius of $R_p = 9r_c$. In the left plot the proximity parameter (section 3.2.1) and in the right plot the distance parameter (section 3.2.2) are used.

batches.
For higher torques ($T_A \geq 10$), the graphs exhibit a step-like pattern, and predator success is very high, except for the combinations $T_A = 10, R_{pp} = 5r_c$ followed by $T_A = 30, R_{pp} = 3r_c$, which is also visible in figure 6c. This can be attributed to the reasoning discussed in section 4.2.2. At a torque of $T_A = 40$, predator success reaches a maximum, with almost all prey particles being eaten.
In figure 8 the number of eating events is plotted over the proximity parameter and average prey-prey distance. All cloud-like prey formations have a very similar survival chance, as well as proximity parameter P and average particle distance D. The survival chance for all other prey states varies highly from simulation to simulation. Generally, more cohesive prey states result in a greater predator success.

4.2.4 Comparison

The highest overall predator success occurs at a high prey-predator interaction range of $R_p = 9r_c$ and a high torque of $T_A = 40$. In these conditions, prey-prey and prey-predator interactions interfere in a way that allows the predator to catch a lot of prey easily. When a predator approaches a cohesive group of prey, the group tends to stay together and remain relatively stationary instead of dispersing. This behavior occurs because each prey individual

is constantly within the interaction range of both the predator and many other prey members. The resulting torques inhibit the prey's ability to effectively evade the predator. That enables the predator to consume the whole group quickly.
The torques where predator success is generally low are $T_\mathrm{A} \in \{0.1, 1\}$. For these torques, the graphs for all R_p values look similar, with larger R_p leading to slightly lower predator success.
For the cloud-like movement pattern at the prey-predator interaction radius $R_\mathrm{p} = 3r_\mathrm{c}$, a prey-prey interaction range of $2r_\mathrm{c}$ results in the lowest predator success. For the other two R_p values, there is no clear correlation between prey-prey interaction radius and predator success for the cloud-like prey movement. At higher torques, where prey movement patterns are more cohesive, predator success is very high across almost all simulations. It is especially high for $R_\mathrm{p} \in \{3r_\mathrm{c}, 9r_\mathrm{c}\}$ and a bit lower for $R_\mathrm{p} = 5r_\mathrm{c}$. In these scenarios, higher prey-prey interaction radii generally lead to higher predator success because the group becomes even more cohesive. The cause of this result is the interference of the cohesion and avoiding mechanics. However, for certain combinations of the prey-prey interaction torque T_A and radius R_pp the group reacts in a coherent way and the predator success reaches a minimum. This combination of parameters is different for all prey-predator interaction radii R_p. For $R_\mathrm{p} = 3r_\mathrm{c}$ the optimal combination is $T_\mathrm{A} = 40$ and $R_\mathrm{pp} = 3r_\mathrm{c}$. For $R_\mathrm{p} = 5r_\mathrm{c}$ it is $R_\mathrm{pp} = 4r_\mathrm{c}$ and $T_\mathrm{A} = 30$ and for $R_\mathrm{p} = 9r_\mathrm{c}$ the two low predator success parameter sets are: $T_\mathrm{A} = 10, R_\mathrm{pp} = 5r_\mathrm{c}$ followed by $T_\mathrm{A} = 30, R_\mathrm{pp} = 3r_\mathrm{c}$. The prey-prey and prey-predator torque interactions' interplay leads to many slightly different prey behaviors, of which only a few seem to yield good results for the prey.

4.3 Dynamics at small interaction radii and stronger torque

In the previous sections, a prey-predator torque of $T_0 = 2T_\mathrm{A}$ was used in the simulations. This section explores the results when the ratio is increased to $T_0 = 3T_\mathrm{A}$, which is expected to enhance the prey's reaction strength towards the predator. A prey-predator interaction radius of $R_\mathrm{p} = 3r_\mathrm{c}$ is used, similar to section 4.2.2. The results are displayed in figures 6d, 13 and 14. In figure 13 the survivors over time for all batches' simulations are shown. For torques of $T_\mathrm{A} \geq 10$, prey survival chances shortly after predator introduction are very low, as previously explained. At low torques of $T_\mathrm{A} \in \{0.1, 1\}$, the survivorship function shows an almost linear decline, following a similar pattern regardless of the interaction radius R_pp. At higher torques ($T_\mathrm{A} \geq 10$), the graphs exhibit a step-like pattern, with generally higher predator success, except for a minimum at $T_\mathrm{A} = 30, R_\mathrm{pp} = 4r_\mathrm{c}$. Overall, the results are consistent with those in previous sections, though the decline of the survivorship function is less pronounced. That means that the predator success is lower overall, which can also be seen in figure 6d.
In figure 14 the number of eating events is plotted over the proximity parameter and average prey-prey distance. There is only a slight increase in predator success for more cohesive prey formations. At high proximity parameters P and large average particle distances D

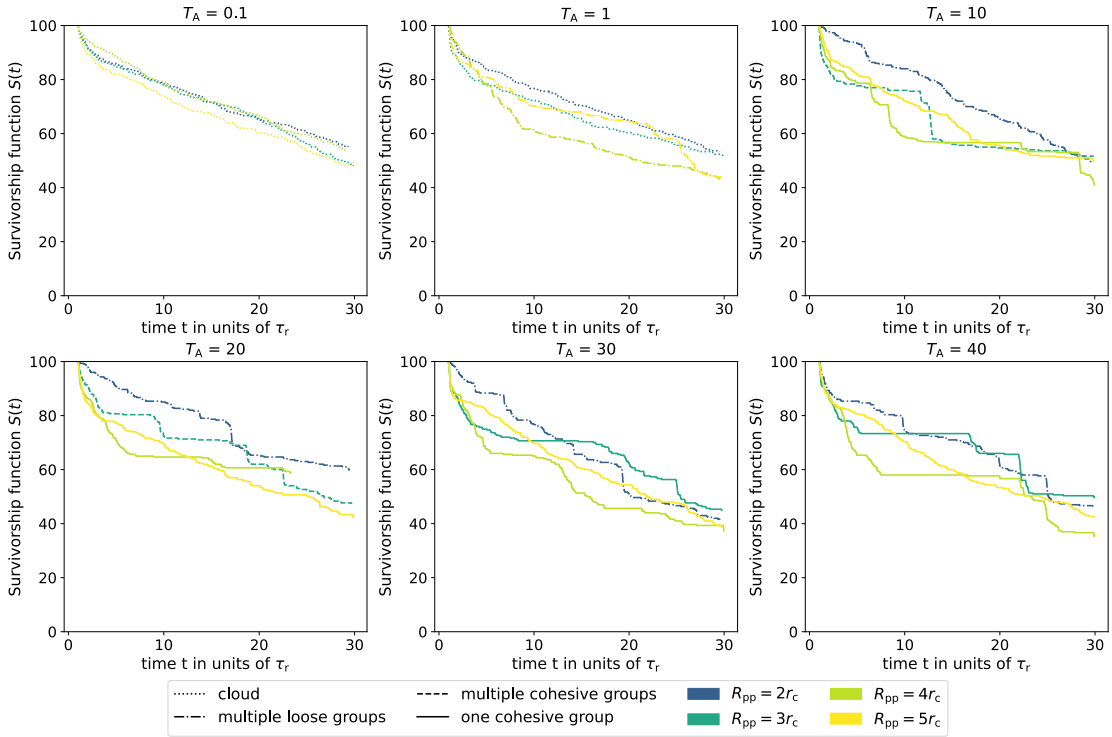


Figure 13: **Survivors over time at $R_p = 3r_c$ and $T_0 = 3T_A$:** This plot shows the number of survivors over time for all 24 simulations for the prey-predator interaction radius of $R_p = 3r_c$ and a torque of $T_0 = 3T_A$, each averaged over three trials. The linestyle shows the prey state explained in section 2.1. A solid line represents the prey moving as one cohesive group; the dashed line means there are multiple cohesive groups; the dashed-dotted line refers to the prey moving in multiple loose groups; and the dotted line indicates there is a cloud-like prey movement. The colors represent the different prey-prey interaction radii R_{pp}.

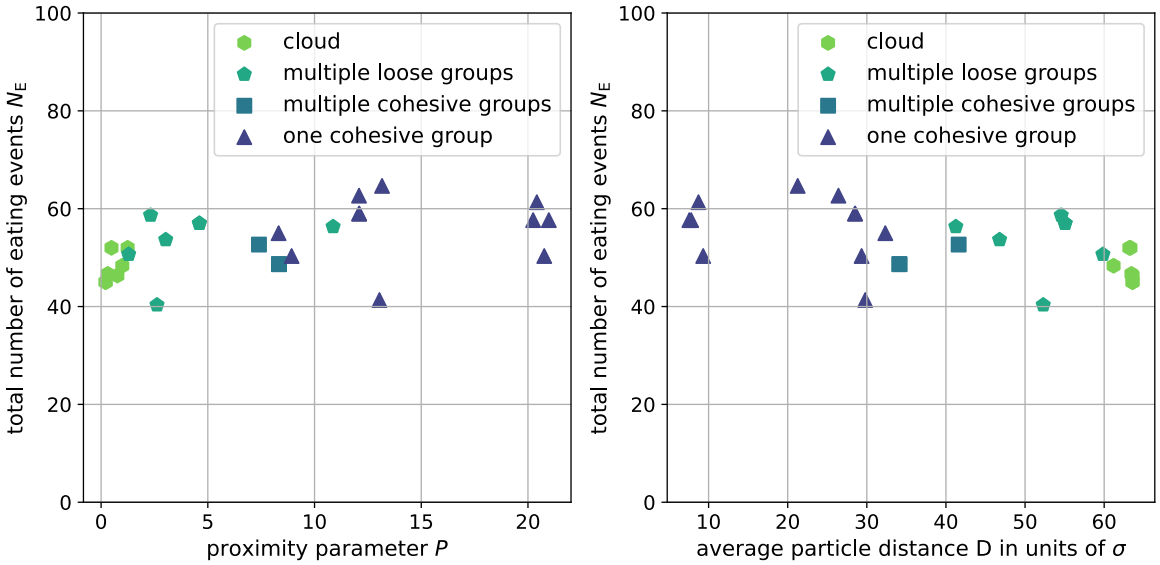


Figure 14: **Number of eating events for prey proximity and distance at $R_\mathbf{p} = 3r_\mathbf{c}$ and $T_0 = 3T_\mathbf{A}$:** This graph shows the correlation between the prey swarms' movement pattern and the number of eaten prey for the prey-predator interaction radius of $R_\mathrm{p} = 3r_\mathrm{c}$ and torque $T_0 = 3T_\mathrm{A}$. In the left plot the proximity parameter (section 3.2.1) and in the right plot the distance parameter (section 3.2.2) are used.

the prey survival chance is lowest. It reaches a maximum at medium P and D, so predator success is lowest when the prey move in more loose groups. The predator success for the larger prey-predator toque ratio is smaller compared to the simulation batches discussed in the sections above.

5 Conclusion and outlook

An extended model of interacting Active Brownian Particles is proposed to study predator-prey interactions, focusing on how collective prey behavior affects survival outcomes.
The dynamics of predator-prey relationships are remarkably intricate; even employing a basic model, the outcomes can vary significantly based on the specific parameters chosen. The simulations revealed complex behaviors and dependencies, particularly highlighting the importance of collective prey behavior on predator success rates.
The simulations show that prey particles in the absence of a predator exhibit four distinct behavior states: cloud-like, multiple loose groups, multiple cohesive groups, and one cohesive group. These states transition based on the prey-prey interaction torque T_{A} and radius R_{pp}. Prey survival rates vary strongly, with the predator's success in catching prey highly depending on the prey's interaction parameters and their resulting collective behavior. When prey exhibit cloud-like movement patterns, predator success is low. The prey move almost independently, each evading the predator on its own, leading to a continuous decline in the number of living prey particles. High predator success occurs when prey moves in cohesive groups. These groups are more easily targeted once within the predator's range, leading to large singular eating events. Large groups also interfere with the prey evasion mechanism
However, the degree of predator success varies significantly across simulations due to the complex interplay between prey-prey and prey-predator interactions. Interestingly, specific combinations of prey-prey interaction parameters for cohesive prey groups resulted in minimal predator success, highlighting a favorable parameter set for prey cohesion and avoidance.
In the future, it would be interesting to increase the number of simulated prey particles in this model, as with increased group size, emergent behavior becomes more pronounced. For instance, with a prey group size of $N = 1000$, the collective behavior of the prey particles would have a more significant impact on the results. This is because larger groups exhibit more complex and realistic interactions, reducing random fluctuations and noise.
Extending the simulation duration would also provide a clearer understanding of the interactions as the number of prey particles diminishes over time. Additionally, examining more varied initial conditions or implementing a longer delay before the appearance of the predator could lead to valuable insights. Another possibility would be to increase the number of predators to observe how the dynamics evolve under these conditions.
While our model offers valuable insights, it operates under certain assumptions, such as simplified two-dimensional movement and idealized interaction forces. Future research could build on this work by exploring three-dimensional environments, incorporating more complex interaction rules, or considering environmental factors such as obstacles or varying terrain.

6 Bibliography

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